

Data Visualization of Bike-Sharing Demand

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1 Business Problem

1.1 Background

Bike-sharing companies face challenges in meeting fluctuating demand for bicycle rentals. Demand can vary by time of day, season, and weather conditions, making it difficult to ensure enough bicycles are available when needed. Underestimating demand can lead to frustrated customers and missed revenue, while overestimating demand can result in unused bicycles and unnecessary operational costs. This project will use the Capital Bikeshare dataset to create a dashboard that serves as a visual aid for understanding these demand patterns. By displaying rental trends and comparisons, the dashboard will help support decisions about bike availability, resource allocation, and maintenance scheduling.

1.2 Business Question

How can a dashboard of historical bike-sharing data help a company understand rental demand in relation to:

1. time of day and season;
2. weather conditions and temperature;
3. working days compared with weekends and holidays; and
4. rental patterns among casual and registered users?

1.3 Business Importance

Understanding when rental demand is higher or lower can help bike-sharing companies prepare for busy periods and schedule maintenance during quieter periods. A dashboard can make these patterns easier to identify and communicate by presenting the data through clear charts and comparisons. This visual aid can support more informed planning, improve customer service, and help reduce inefficient use of resources.

1.4 Analytical Objectives

This project will:

1. summarize daily and hourly bike rental counts;

2. identify peak rental hours and seasonal demand patterns;
3. examine how rental counts vary with temperature and weather conditions;
4. compare demand on working days with weekends and holidays;
5. compare rental patterns between casual and registered users; and
6. create a dashboard that clearly communicates these patterns to support decisions about bike availability, resource allocation, and maintenance scheduling.